Meta-Learning Based Signal Demodulation Using Few Pilot Samples

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***Abstract*—In Internet-of-Things (IoT) communication systems, devices transmit short packets with very limited pilot symbols, which makes conventional demodulation techniques ineffective due to insufficient training data. In such scenarios, accurate estimation of the channel and hardware impairments becomes challenging. This work implements a meta-learning-based demodulation framework using Model-Agnostic Meta-Learning (MAML), enabling rapid adaptation to new channel conditions using only a few pilot samples. By leveraging pilot data from multiple devices during meta-training, the model learns a parameter initialization that can be efficiently adapted to unseen devices. The framework is evaluated using both a simplified toy model and a realistic scenario incorporating Rayleigh fading and nonlinear amplifier distortion. The results demonstrate that the proposed approach significantly outperforms conventional learning methods in low-data regimes.**

In addition, the proposed framework incorporates a nonlinear device-aware signal model to capture practical hardware distortions, along with a four-dimensional feature representation combining Cartesian and polar components to improve signal characterization. A residual neural network architecture is employed to enhance learning capability and stability during adaptation. Furthermore, an online continual learning extension is introduced to handle sequentially arriving devices, enabling the model to retain prior knowledge while adapting to new conditions.

***Index Terms*—Meta-learning, demodulation, wireless communication, MAML.**

I. Introduction

Wireless communication systems fundamentally rely on the ability of a receiver to accurately recover transmitted symbols from received signals. Classical demodulation techniques assume either perfect knowledge of the channel or the availability of sufficient pilot symbols to estimate the channel state information. However, in modern IoT systems, devices operate under strict energy and latency constraints, transmitting short packets that contain only a few pilot symbols. As highlighted in the original work , this limited pilot availability makes accurate channel estimation infeasible, particularly in the presence of nonlinear hardware distortions and unknown channel conditions.

In such environments, demodulation can be formulated as a classification problem in which the received signal is mapped to one of the possible transmitted symbols. Traditional supervised learning approaches require large amounts of labeled data and therefore fail in few-shot scenarios. To overcome this limitation, meta-learning provides a framework that leverages prior experience across multiple related tasks to enable rapid adaptation to new tasks using minimal data. In this work, a meta-learning-based demodulator is implemented that learns a generalizable initialization capable of adapting to new devices using only a small number of pilot symbols.

II. System Model and Problem Formulation

A. System Model

We consider a wireless communication system consisting of multiple IoT devices communicating with a base station. Each device transmits symbols selected from a finite constellation set $\mathcal{S}$, determined by the modulation scheme. Let $s_k \in \mathcal{S}$ denote the symbol transmitted by the k-th device and y_k denote the corresponding received signal at the base station.

In practical systems, the transmitted signal does not propagate directly to the receiver in its original form. Instead, it undergoes several transformations that collectively define the end-to-end channel. One of the primary sources of distortion arises from the transmitter's power amplifier, which introduces nonlinear effects due to hardware limitations. As a result, the transmitted symbol s_k is transformed into a distorted signal x_k, which can be expressed as

$$x_k = g_k(s_k), \tag{1}$$

where $g_k(\cdot)$ represents the nonlinear transfer function associated with the k-th device.

A commonly used model for this nonlinearity assumes amplitude distortion while preserving the phase of the signal. Under this model, the distorted signal can be written as

$$x_k = \frac{\alpha_k |s_k|}{1 + \beta_k |s_k|^2} \exp\left(j \angle s_k\right), \tag{2}$$

where $|s_k|$ and $\angle s_k$ denote the magnitude and phase of the symbol, respectively, and α_k and β_k are parameters characterizing the amplifier. This model captures the saturation behavior of practical amplifiers, where the output amplitude compresses as the input magnitude increases.

Following the nonlinear distortion, the signal propagates through a wireless channel, which introduces fading effects due to multipath propagation. The channel is modeled as a complex gain h_k, which remains constant over the duration of transmission for a given device. Additionally, the received signal is corrupted by additive noise, typically modeled as complex Gaussian noise $z_k \sim \mathcal{CN}(0, N_0)$. Combining these effects, the received signal can be expressed as

$$y_k = h_k x_k + z_k. \tag{3}$$

Substituting the expression for x_k, the end-to-end channel model becomes

$$y_k = h_k g_k(s_k) + z_k. \tag{4}$$

The receiver does not have access to the exact channel gain h_k or the nonlinear function $g_k(\cdot)$, making direct inversion of the channel infeasible. Therefore, the objective is to estimate the transmitted symbol s_k from the received signal y_k with high probability. This is formally expressed as

$$\hat{s} = \arg\max_{s \in \mathcal{S}} p(s \mid y). \tag{5}$$

Instead of explicitly estimating the channel, a data-driven approach is adopted in which the demodulator is modeled as a conditional probability distribution $p_\theta(s \mid y)$, parameterized by a neural network with parameters θ.

B. Specific Channel Models

The system model described above highlights a fundamental challenge in wireless communication systems with limited pilot data. In conventional receivers, demodulation is typically performed by first estimating the channel and then applying an optimal detection rule. However, such approaches rely heavily on the availability of sufficient pilot symbols to accurately estimate the channel parameters. In IoT scenarios, where only a small number of pilot symbols is available, this approach becomes unreliable.

From a learning perspective, the demodulation task can be viewed as learning a mapping from received signals y to transmitted symbols s . A straightforward approach would be to train a model using pilot data from a single device. However, due to the limited number of pilots, the model would either underfit or overfit, resulting in poor generalization.

To address this limitation, the problem is reformulated within the meta-learning framework. Instead of learning a single task, the system considers multiple related tasks corresponding to different devices. Each device is characterized by its own channel realization and hardware distortion, which together define a unique input-output relationship. Although these relationships differ across devices, they share a common underlying structure, since all devices operate under the same communication principles.

Let us denote by $k$the index of a device. For each device k, we define a dataset consisting of pairs of transmitted symbols and received signals:

$$D_k = \{(s_k^{(n)}, y_k^{(n)})\}_{n=1}^{N}. \tag{6}$$

These datasets are collected from multiple devices, forming a meta-training dataset

$$D = \{D_k\}_{k=1}^{K}. \tag{7}$$

Each dataset D_k represents a separate learning task. The goal of meta-learning is to utilize these tasks to learn a model that can generalize across devices and adapt quickly to a new device using only a few pilot samples.

For a new device (meta-test device), only a small set of pilot samples is available:

$$D_T = \{(s^{(n)}, y^{(n)})\}_{n=1}^{P}. \tag{8}$$

The objective is to learn a demodulator $p_\theta(s \mid y)$ such that, after observing the small dataset D_T, the model can accurately predict transmitted symbols for new received signals. This leads to the formulation of the following problem: find a parameter vector θ such that the model can be efficiently adapted to minimize the loss on the target device after observing only a few pilot samples.

To evaluate the proposed framework, two different channel models are considered.

1) Toy Model: In the first setting, a simplified toy model is used to provide intuition about the behavior of the meta-learning algorithm. The modulation scheme is 4-PAM with symbol set

$$\mathcal{S} = \{-3, -1, 1, 3\}. \tag{9}$$

The channel is modeled as binary fading, where the channel gain takes values

$$h \in \{+1, -1\}. \tag{10}$$

The received signal is given by

$$y = h \cdot s + n, \tag{11}$$

where $n \sim \mathcal{N}(0, \sigma^2)$.is additive Gaussian noise

The noise variance is determined by the signal-to-noise ratio (SNR) as

$$\sigma^2 = \frac{\mathbb{E}[s^2]}{10^{\text{SNR}/10}}. \tag{12}$$

Even in this simplified setting, the meta-learned initialization enables rapid error reduction with very few pilots compared to conventional approaches.

2) Realistic Channel Model: In the second setting, a more realistic communication scenario is considered, involving 16-QAM modulation, Rayleigh fading, and nonlinear amplifier distortion. The transmitted symbols are drawn from a complex constellation

$$s \in \{-3, -1, 1, 3\} + j\{-3, -1, 1, 3\}. \tag{13}$$

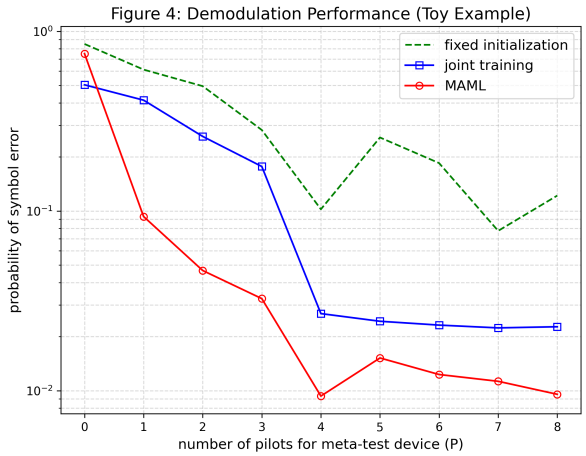

Fig. 1. Symbol error rate versus number of pilot samples for the toy model.

The effectiveness of learning in this simplified setting is illustrated in Figure 1, which shows the symbol error rate as a function of the number of pilot samples. Even in this basic scenario, the meta-learned initialization enables rapid error reduction with very few pilots compared to conventional approaches.

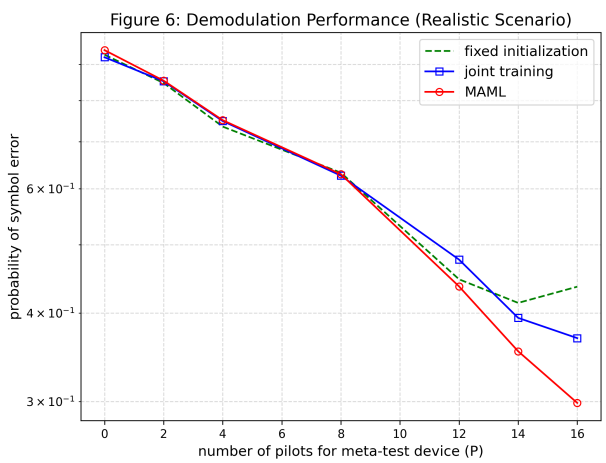

Fig. 2. Symbol error rate versus number of pilot samples for the realistic channel model.

The performance under these realistic channel conditions is shown in Fig. 2, where the symbol error rate is plotted against the number of pilot samples.

The nonlinear distortion is modeled using the same amplitude-dependent function, and the channel gain is modeled as a complex Gaussian random variable

$$h \sim \mathcal{CN}(0, 1). \tag{14}$$

The received signal is expressed as

$$y = hx + z, \tag{15}$$

where $z \sim \mathcal{CN}(0, \sigma^2)$.

The noise variance is computed as

$$\sigma^2 = \frac{\mathbb{E}[|s|^2]}{10^{\mathrm{SNR}/10}}. \tag{16}$$

Despite increased channel complexity, the meta-learning approach maintains superior performance, demonstrating robustness to fading and nonlinear distortion.

III. Learning-Based Demodulation

Given the system model and the meta-learning formulation, the demodulation problem is approached using a neural network that approximates the conditional probability distribution $p_\theta(s \mid y)$. This formulation treats demodulation as a multi-class classification problem, where each class corresponds to a symbol in the constellation set $\mathcal{S}$.

The neural network takes the received signal y as input and produces a vector of scores, one for each possible symbol. These scores are converted into probabilities using the softmax function:

$$p_\theta(s = k \mid y) = \frac{\exp\left(f_\theta(y)_k\right)}{\sum_j \exp\left(f_\theta(y)_j\right)}. \tag{17}$$

The model is trained by minimizing the cross-entropy loss, which measures the discrepancy between the predicted probabilities and the true transmitted symbols:

$$\mathcal{L}(\theta) = -\sum_i \log p_\theta(s_i \mid y_i). \tag{18}$$

This loss function encourages the model to assign high probability to the correct symbol and low probability to incorrect symbols. However, in the context of few-shot learning, minimizing this loss directly for a single device is insufficient, as the amount of training data is extremely limited. This motivates the use of meta-learning to improve generalization.

The behavior of the learned conditional distribution is illustrated in Fig. 3. Initially, the model produces smooth and overlapping probability distributions across symbols, reflecting uncertainty in decision regions. After adaptation with a few pilot samples, these distributions become sharply concentrated around the correct symbols, indicating confident and accurate demodulation.

IV. Neural Network Architecture

The neural network architecture is designed to approximate the nonlinear mapping from received signals to transmitted symbols. The structure of the network depends on the complexity of the channel model.

In the toy example, a relatively simple network is sufficient due to the simplicity of the channel. The network consists of an input layer, a single hidden layer, and an output layer. The hidden layer applies a nonlinear transformation to the input:

$$h_1 = \tanh(W_1 y + b_1), \tag{19}$$

followed by a linear transformation to produce the output logits:

$$z = W_2 h_1 + b_2. \tag{20}$$

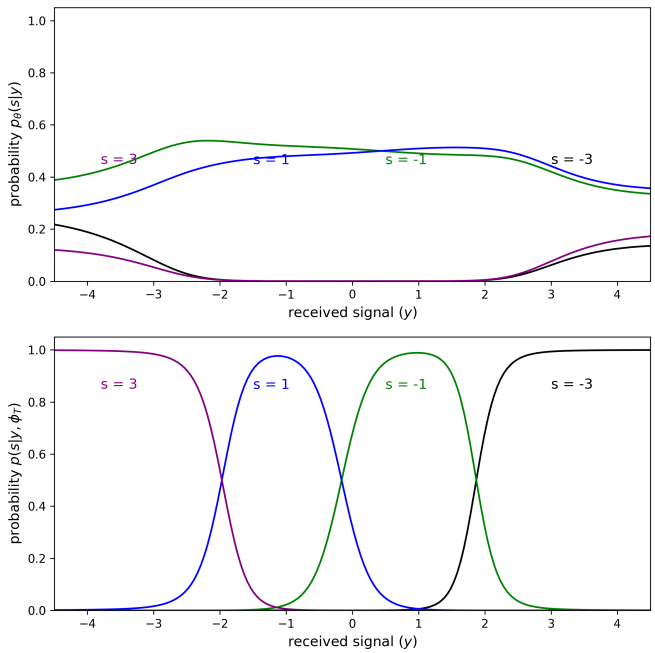

Fig. 3. Conditional probability distributions $p_\theta(s \mid y)$ for different transmitted symbols. *Top:* Before adaptation, the distributions are smooth and overlapping, indicating uncertainty in decision regions. *Bottom:* After adaptation with a few pilot samples, the distributions become sharply peaked around the correct symbols, enabling accurate demodulation.

In the realistic scenario, the channel includes nonlinear distortion and fading, requiring a more expressive model. A deeper network is used with two hidden layers, each followed by a ReLU activation function:

$$h_1 = \mathrm{ReLU}(W_1 y + b_1), \tag{21}$$

$$h_2 = \mathrm{ReLU}(W_2 h_1 + b_2), \tag{22}$$

$$z = W_3 h_2 + b_3. \tag{23}$$

The increased depth allows the network to capture more complex relationships between the received signal and the transmitted symbol.

V. Meta-Learning Framework

The central idea of meta-learning is to learn a parameter initialization that enables rapid adaptation to new tasks. In this context, each task corresponds to a different device with its own channel and distortion characteristics.

The training process consists of two nested optimization loops. The inner loop corresponds to task-specific adaptation, while the outer loop corresponds to meta-optimization across tasks.

For a given device k, the available dataset D_k is divided into two subsets: a training subset D_k^{tr} and a testing subset D_k^{te}. The training subset is used to compute an adapted parameter vector, while the testing subset is used to evaluate the performance of the adapted model.

The adaptation step is performed using gradient descent:

$$\theta'_k = \theta - \eta \nabla_\theta \mathcal{L}_{D_k^{\mathrm{tr}}}(\theta), \tag{24}$$

where η is the learning rate and $\mathcal{L}_{D_k^{\mathrm{tr}}}(\theta)$ is the loss computed on the training subset.

The meta-learning objective is to minimize the loss of the adapted parameters across all tasks:

$$\min_\theta \sum_{k=1}^{K} \mathcal{L}_{D_k^{\mathrm{te}}}(\theta'_k). \tag{25}$$

This objective ensures that the initial parameter θ is chosen such that a small update based on limited data leads to good performance.

The meta-update is performed using gradient descent:

$$\theta \leftarrow \theta - \kappa \nabla_\theta \sum_{k=1}^{K} \mathcal{L}_{D_k^{\mathrm{te}}}(\theta'_k), \tag{26}$$

where κ is the meta-learning rate.

During deployment, the learned initialization is adapted to a new device using the available pilot samples:

$$\theta' = \theta - \eta \nabla_\theta \mathcal{L}_{D_T}(\theta). \tag{27}$$

This adapted parameter is then used for demodulation of subsequent symbols.

VI. Performance Metric

The performance of the demodulator is evaluated using the Symbol Error Rate (SER), which measures the fraction of incorrectly predicted symbols:

$$\mathrm{SER} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}(\hat{s}_i \neq s_i). \tag{28}$$

This metric provides a direct measure of demodulation accuracy and is used to compare different learning approaches.

VII. Results and Analysis

The performance of the proposed method is evaluated for both the toy example and the realistic scenario. In the toy example, the results demonstrate that the meta-learned model achieves a rapid decrease in error as the number of pilot samples increases. The model is able to reach very low error rates with only a few pilots, while joint training and fixed initialization perform significantly worse. This behavior indicates that the learned initialization is well-positioned in the parameter space such that only minimal adaptation is required to align with a new device. Unlike conventional approaches that attempt to learn a single global model, the meta-learning framework explicitly optimizes for adaptability, enabling efficient learning even in extremely low-data regimes. As previously illustrated in Figure 4, the error curve corresponding to the meta-learning approach drops sharply within the first few pilot samples, highlighting its effectiveness in few-shot settings.

The probability distributions before and after adaptation illustrate that the meta-learned model initially maintains a

balanced representation of the decision regions, which allows it to quickly specialize to the channel conditions of a new device. Before adaptation, the conditional distributions over symbols are smooth and overlapping, reflecting uncertainty due to unknown channel parameters. After observing a small number of pilot samples, these distributions become sharply peaked around the correct symbols, indicating confident predictions aligned with the underlying channel. This transition corresponds to a rapid reshaping of decision boundaries in the feature space, where the model shifts from a general-purpose representation to a task-specific configuration. The ability to perform such a transformation with limited data is a direct consequence of the meta-training process, which encodes shared structural information across tasks.

In addition to improved accuracy, the meta-learning approach exhibits more stable convergence behavior compared to baseline methods. Fixed initialization lacks the flexibility required to adapt effectively, resulting in persistently high error rates. Joint training, while more flexible, requires a significantly larger number of samples to achieve comparable performance, as it does not explicitly account for task variability during training. Consequently, both methods are less suitable for scenarios where only a small number of pilot symbols is available.

In the realistic scenario, which includes Rayleigh fading and nonlinear distortion, the performance gap between the proposed method and baseline approaches remains significant. Although more pilot samples are required compared to the toy example, the meta-learning approach continues to demonstrate superior adaptability and robustness. As shown in Figure 6, all methods exhibit a gradual decrease in error with increasing pilot samples; however, the proposed method consistently achieves lower symbol error rates across all regimes. The advantage is particularly pronounced in the low-pilot region, where conventional methods struggle to capture the complex nonlinear relationships introduced by channel fading and hardware impairments.

The increased difficulty in the realistic scenario arises from the combined effects of stochastic channel variations and nonlinear signal distortion, which significantly complicate the mapping from received signals to transmitted symbols. Despite these challenges, the meta-learned model is able to leverage prior knowledge acquired during training to quickly adapt to new channel conditions. This demonstrates that the framework effectively captures invariant features across tasks while retaining the flexibility to adjust to device-specific characteristics.

Overall, the results confirm that the proposed meta-learning-based demodulation approach provides a substantial improvement over traditional learning strategies in both simplified and realistic environments. The ability to achieve low error rates with very few pilot samples makes this approach particularly suitable for practical IoT communication systems, where data is limited and channel conditions are highly variable.

VIII. Discussion

The results indicate that the proposed framework effectively captures the underlying structure shared across different devices. Instead of explicitly estimating the channel or the nonlinear distortion, the model learns a direct mapping from received signals to transmitted symbols. This enables efficient adaptation using very limited data. The ability to generalize across devices and quickly specialize to new conditions highlights the effectiveness of meta-learning in communication systems with constrained training data.

IX. Novelty

A. Nonlinear Device-Aware Signal Model

In practical wireless communication systems, transmitted signals are affected not only by channel fading but also by hardware-induced nonlinearities. To accurately model this behavior, the transmitted symbol $s \in \mathcal{S}_{\text{16QAM}}$ is passed through a nonlinear transformation defined as

$$x = \frac{4|s|}{1+\beta|s|^2} e^{j\angle s} \tag{29}$$

where $|s|$ and $\angle s$ denote the magnitude and phase of the transmitted symbol, respectively, and β is a device-dependent nonlinearity coefficient. This formulation models amplitude compression effects typically observed in power amplifiers, where higher magnitude symbols experience stronger distortion.

The received signal is then expressed as

$$y = hx + n \tag{30}$$

where $h \sim \mathcal{CN}(0,1)$ represents the complex channel coefficient capturing Rayleigh fading, and $n \sim \mathcal{CN}(0,\sigma^2)$ denotes additive white Gaussian noise. The noise variance is determined from the signal-to-noise ratio (SNR) as

$$\sigma^2 = \frac{\mathbb{E}[|s|^2]}{10^{\text{SNR}_{\text{dB}}/10}} \tag{31}$$

This formulation introduces variability across devices through both h and β, thereby defining a distribution of tasks suitable for meta-learning. Each device corresponds to a distinct realization of channel and hardware conditions, requiring rapid adaptation.

B. Four-Dimensional Feature Representation

Conventional demodulation techniques typically rely on the in-phase and quadrature components of the received signal. However, such representations are insufficient to fully capture distortions introduced by nonlinear effects.

To address this limitation, a four-dimensional feature vector is constructed as

$$x = [R(y), I(y), |y|, \angle y] \tag{32}$$

where $R(y)$ and $I(y)$ represent the real and imaginary parts of the received signal, while $|y|$ and $\angle y$ denote its magnitude and phase.

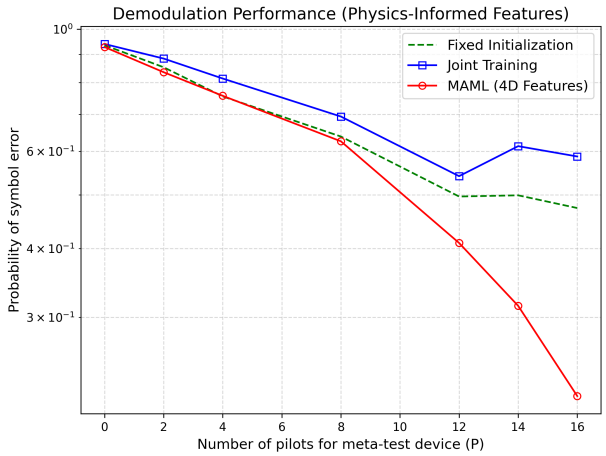

Fig. 4. Symbol Demodulation performance comparison using physics-informed 4D feature representation.

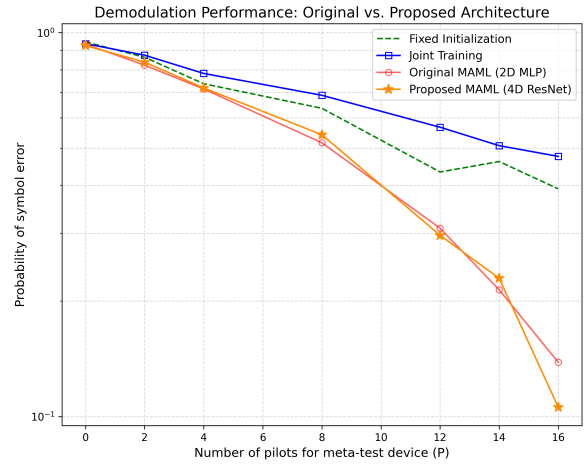

Fig. 5. Symbol Comparison of demodulation performance between original MAML (2D MLP) and proposed MAML (4D ResNet architecture).

To ensure numerical stability and prevent gradient explosion during training, normalization is applied to the polar components as

$$|y|_{\text{norm}} = \frac{|y|}{4}, \quad \angle y_{\text{norm}} = \frac{\angle y}{\pi} \tag{33}$$

The resulting feature vector combines both Cartesian and polar representations, enabling the model to simultaneously learn amplitude distortions, phase shifts, and linear signal structure. This hybrid representation significantly enhances feature separability in the presence of nonlinear channel effects.

Fig. 4 illustrates the impact of the proposed four-dimensional feature representation on demodulation performance as a function of the number of pilot symbols. It can be observed that the MAML-based approach utilizing 4D features achieves a consistently lower symbol error rate compared to both fixed initialization and joint training methods. The performance gap becomes increasingly significant as the number of pilots increases, indicating that the enriched feature space improves the model's ability to exploit limited adaptation data. This validates that incorporating both Cartesian and polar components enables better characterization of nonlinear distortions and phase variations, thereby enhancing feature separability and improving generalization across device conditions.

C. Residual Neural Network Architecture

To effectively learn from the enriched feature representation, a deep neural architecture incorporating residual connections is employed. The forward propagation of the network is defined as

$$h_1 = \text{ReLU}(W_1 x + b_1) \tag{34}$$

$$h_2 = \text{ReLU}(W_2 h_1 + b_2) \tag{35}$$

$$h_3 = \text{ReLU}(W_3 h_2 + b_3 + h_1) \tag{36}$$

$$\hat{y} = W_4 h_3 + b_4 \tag{37}$$

where W_i and b_i denote the weight matrices and bias vectors of the respective layers. The activation function used is the rectified linear unit (ReLU).

The key architectural feature is the residual connection introduced in (8), where the output of the first layer h_1 is added to the transformation of deeper layers. This design facilitates improved gradient propagation and mitigates the vanishing gradient problem, which is particularly critical in meta-learning frameworks involving higher-order derivatives.

Furthermore, the residual structure allows the network to preserve low-level representations while simultaneously learning higher-level abstractions, improving robustness under varying channel and hardware conditions.

Fig. 5 compares the demodulation performance of the original MAML implementation based on a 2D MLP with the proposed architecture incorporating 4D features and residual connections. The proposed model demonstrates a clear improvement in symbol error rate across all pilot regimes, with particularly significant gains at higher pilot counts. This improvement can be attributed to the enhanced representational capacity of the residual network and its ability to effectively utilize the richer input features. The residual connections facilitate deeper feature learning while maintaining stable gradient propagation, enabling the model to better capture nonlinear relationships present in the received signal.

D. Meta-Learning Formulation

The proposed system adopts a Model-Agnostic Meta-Learning (MAML) framework to enable rapid adaptation to new devices with limited pilot data.

Let θ denote the model parameters. For each device (task), a support set D_s and a query set D_q are defined.

1) Inner Loop Adaptation

The model parameters are adapted using the support set as

$$\theta' = \theta - \alpha \nabla_\theta L_{(D_s)}(\theta) \tag{38}$$

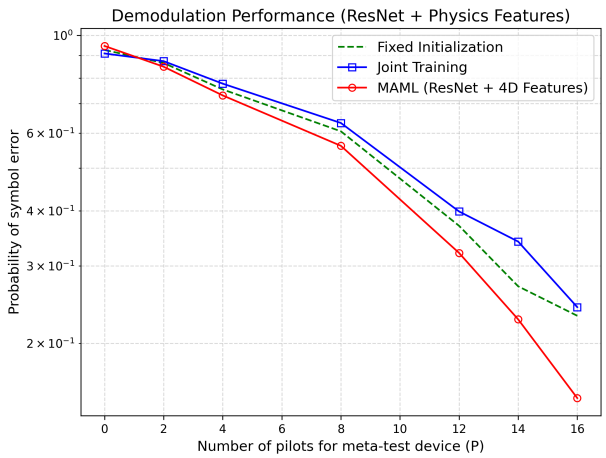

Fig. 6. Symbol Demodulation performance of the proposed model combining residual architecture and physics-informed features.

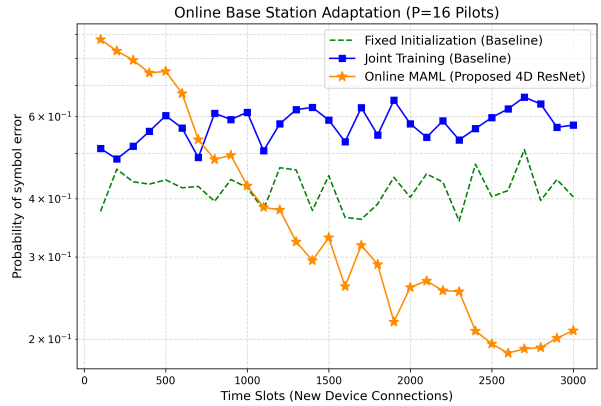

Fig. 7. Symbol Online adaptation performance over time for sequentially arriving devices using the proposed MAML framework.

where α is the inner-loop learning rate and $L_{(D_s)}$ is the cross-entropy loss computed on pilot samples.

This step represents fast adaptation to a specific device using a small number of labeled samples.

2) Outer Loop Optimization

The meta-objective is defined as

$$\min_{\theta} \sum_{k} L_{(D_q^{(k)})}(\theta'_k) \tag{39}$$

where θ'_k denotes the adapted parameters for the k-th device. The objective ensures that the initial parameters θ are optimized such that minimal adaptation is required for unseen devices.

This formulation enables the model to learn a parameter initialization that generalizes across a distribution of device conditions.

E. Fast Adaptation Mechanism

A key property of the proposed system is its ability to achieve effective adaptation within a small number of gradient steps. The iterative update during adaptation is given by

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \nabla_{\theta} L(\theta^{(t)}) \tag{40}$$

where t denotes the adaptation step.

Due to the meta-learned initialization and enhanced architecture, the model converges rapidly, typically requiring only a few update steps. This significantly reduces computational overhead and enables real-time deployment in dynamic communication environments.

Fig. 6 presents the overall performance of the proposed system integrating both physics-informed feature representation and residual network architecture within the MAML framework. The model consistently outperforms conventional baselines, achieving the lowest symbol error rate across all pilot configurations. Notably, the performance gain becomes more pronounced with increasing pilot samples, demonstrating the model's ability to rapidly adapt using limited data. This behavior confirms that the combination of improved feature design and architectural enhancements leads to more efficient gradient-based adaptation, thereby validating the effectiveness of the proposed approach in practical demodulation scenarios.

F. Online Continual Learning Framework

To further extend the applicability of the model, an online learning framework is introduced in which devices arrive sequentially over time.

At each time step t, a new device characterized by parameters (h_t, β_t) is observed. The model is updated using a replay buffer containing previously encountered devices.

The meta-objective in this setting is defined as

$$L_{\text{meta}} = \mathbb{E}_{(h,\beta)\sim B}\left[L_{(D_q)}(\theta')\right] \tag{41}$$

where B denotes the replay buffer.

This approach enables continuous improvement of the model while retaining knowledge of past devices, thereby addressing the challenges of non-stationary environments.

Fig. 7 demonstrates the effectiveness of the proposed online continual learning framework in adapting to sequentially arriving devices over time. While baseline methods such as fixed initialization and joint training exhibit relatively stable but higher error rates, the proposed online MAML approach shows a steady decrease in symbol error rate as more devices are observed. This trend indicates that the model successfully accumulates knowledge from past tasks through the replay buffer and continuously refines its initialization. The results highlight the ability of the framework to operate in non-stationary environments, maintaining improved performance without catastrophic forgetting.

G. Training Objective and Performance Metric

The proposed model is trained as a multi-class classification problem over the 16-QAM constellation, where each transmitted symbol corresponds to one of 16 discrete classes. The

training objective is defined using the categorical cross-entropy loss:

$$L = -\sum_{c=1}^{16} y_c \log(\hat{y}_c) \tag{42}$$

where $y_c \in \{0, 1\}$ is the one-hot encoded ground truth label indicating the transmitted symbol class, and $\hat{y}_c$ represents the predicted posterior probability for class c, obtained from the softmax output of the neural network. The softmax function is given by

$$\hat{y}_c = \frac{\exp(z_c)}{\sum_{k=1}^{16} \exp(z_k)} \tag{43}$$

where z_c denotes the logit corresponding to class c.

From a probabilistic perspective, minimizing the cross-entropy loss is equivalent to maximizing the log-likelihood of the observed data under the model distribution. Specifically, (14) corresponds to the negative log-likelihood of the correct class:

$$L = -\log P(y \mid x; \theta) \tag{44}$$

Thus, the training procedure encourages the model to assign high probability to the correct transmitted symbol while suppressing incorrect classes.

In the context of communication systems, this formulation can be interpreted as learning an approximation to the optimal maximum a posteriori (MAP) decision rule:

$$\hat{s} = \arg\max_{s \in S} P(s \mid y) \tag{45}$$

where S denotes the constellation set. The neural network effectively learns this decision boundary in a data-driven manner, accounting for nonlinear distortion and channel effects.

H. Connection to Symbol Error Rate

While the model is trained using cross-entropy loss, the primary performance metric of interest is the symbol error rate (SER), defined as

$$\text{SER} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}(\hat{y}_i \neq y_i) \tag{46}$$

where N is the total number of transmitted symbols, y_i is the true label, and $\hat{y}_i = \arg\max_c \hat{y}_{(i,c)}$ is the predicted class label. The indicator function $\mathbf{1}(\cdot)$ evaluates to 1 when a misclassification occurs and 0 otherwise.

The SER provides a direct measure of demodulation performance and is analogous to the probability of symbol error in classical communication theory. Unlike cross-entropy, which evaluates probabilistic confidence, SER strictly measures decision accuracy.

1) Relationship Between Cross-Entropy and SER: Although SER is the ultimate evaluation metric, it is non-differentiable due to the discrete argmax operation in (15), making it unsuitable for gradient-based optimization. Therefore, cross-entropy is employed as a surrogate loss function.

Minimizing cross-entropy leads to improved SER due to the following reasons. First, cross-entropy penalizes incorrect predictions more severely when the model is confidently wrong, thereby encouraging sharper decision boundaries. Second, it ensures that the predicted probability distribution becomes increasingly concentrated around the correct class, which directly improves classification accuracy.

Formally, as the predicted probability for the correct class approaches unity:

$$\hat{y}_{\text{true}} \to 1 \Rightarrow L \to 0 \tag{47}$$

which implies that the probability of misclassification decreases. Consequently, minimizing (14) indirectly minimizes the SER defined in (15).

2) Interpretation Under Noisy and Nonlinear Channels: In the presence of nonlinear distortion and channel fading, the received signal distribution becomes highly complex and deviates from standard Gaussian assumptions. Traditional demodulators rely on analytical decision boundaries derived under simplified models, which may not hold in such scenarios.

By contrast, the proposed learning-based approach optimizes (14) over empirical data generated from the true system model:

$$y = h \cdot \frac{4|s|}{1 + \beta|s|^2} e^{j\angle s} + n$$

This enables the model to implicitly learn optimal decision regions that account for both nonlinear and stochastic effects. As a result, the learned classifier approximates the true conditional distribution $P(s \mid y)$, leading to improved SER performance even under severe distortions.

3) Task-Specific Adaptation and SER Improvement: Within the meta-learning framework, the training objective is applied both before and after adaptation. During the inner loop, cross-entropy loss is minimized over pilot samples to obtain task-specific parameters. During the outer loop, the loss is evaluated on unseen query samples to ensure generalization.

This two-level optimization ensures that the model not only minimizes classification error but also learns to adapt rapidly to new device conditions, thereby maintaining low SER across a wide range of channel and hardware variations.

X. Conclusion

This work presents a comprehensive meta-learning-based framework for signal demodulation in wireless communication systems operating under limited pilot data conditions. By formulating demodulation as a few-shot classification problem and leveraging the Model-Agnostic Meta-Learning (MAML) paradigm, the proposed approach enables rapid adaptation to unseen devices with minimal training samples.

In addition to the baseline meta-learning formulation, several key enhancements are introduced to improve performance in realistic scenarios. A nonlinear device-aware signal model is incorporated to capture practical hardware impairments, while a physics-informed four-dimensional feature representation effectively encodes both Cartesian and polar characteristics of the received signal. Furthermore, the integration of a residual neural network architecture improves gradient flow and enhances representational capacity, enabling robust learning under complex channel conditions.

The experimental results across both toy and realistic channel models demonstrate that the proposed method consistently achieves lower symbol error rates compared to conventional approaches, particularly in low-pilot regimes. The framework not only improves accuracy but also exhibits stable and efficient adaptation behavior, making it highly suitable for IoT and next-generation communication systems where data availability is constrained.

Moreover, the extension to an online continual learning framework highlights the capability of the model to adapt in dynamic environments while retaining knowledge of previously encountered devices. This addresses practical challenges such as non-stationarity and device heterogeneity in real-world deployments.

Overall, the proposed approach establishes that combining meta-learning with domain-aware feature engineering and architectural improvements can significantly enhance demodulation performance in challenging wireless environments. Future work may focus on extending this framework to multi-user scenarios, higher-order modulation schemes, and real-world hardware implementations.

XI. References

[1] S. Park, H. Jang, O. Simeone, and J. Kang, "Learning How to Demodulate from Few Pilots via Meta-Learning" 2019. [Online]. Available: https://github.com/ParthKhiriya/demodulator-meta-learning/blob/main/papers/1903.02184v1.pdf

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